

Coulomb's Law

1 Definition:

The force on a Charge q_1 due to charge q_2 .
Parallels *Newton's Law of Universal Gravitation*.

$$\vec{F}_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

ϵ_0 : permittivity constant = $8.85 \times 10^{-12} \frac{\text{C}^2}{\text{Nm}^2}$

No negative sign $\implies$ like charges repel, opposites attract